

# Jayachandra Sarika

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Data Analyst with 2 years of experience designing, building, and deploying predictive models, data pipelines, and analytical solutions that support operational and strategic decision making. Proficient in Python (pandas, NumPy, scikit-learn), SQL, and feature engineering with hands-on experience conducting exploratory data analysis, geospatial and temporal modeling, and building scalable data workflows. Strong background in model development, experimentation, and deployment with exposure to cloud analytics platforms, version control (GitHub), and Agile practices. Demonstrated ability to communicate complex insights clearly to technical and non-technical stakeholders, collaborate across cross-functional teams, and deliver data-driven solutions aligned with business goals. Familiar with energy sector concepts and eager to apply analytical expertise to insurance and oil & gas domains.

## EDUCATION

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| <b>Master of Science, Computer Science</b>                      | 2024 - 2026      |
| <i>University of Houston, Houston TX</i>                        | GPA: 3.92 / 4.00 |
| <b>Bachelor of Technology, Computer Science and Engineering</b> | 2020 - 2024      |
| <i>Vellore Institute of Technology, Vellore India</i>           | GPA: 3.64 / 4.00 |

## TECHNICAL SKILLS

**Programming & Data Science:** Python (pandas, NumPy, scikit-learn, matplotlib, seaborn), R, SQL, Statistical Modeling, Machine Learning, Feature Engineering, Model Evaluation

**Data Engineering & Pipelines:** SQL (queries, joins, stored procedures, data extraction), Data Pipeline Development, ETL Workflows, Database Design, Data Quality Assurance, Automation (scripting, workflows)

**Exploratory Data Analysis:** Data Profiling, Missing Data Handling, Outlier Detection, Variance Analysis, Geospatial Analysis, Temporal Feature Engineering, Entity Matching Methods

**Tools & Platforms:** Power BI, Tableau, Excel (advanced: pivot tables, modeling, VBA), Version Control (Git, GitHub), Cloud Platforms (AWS, Azure, Databricks exposure), Agile/SDLC Practices

**Business & Communication:** Stakeholder Collaboration, Cross-Functional Teamwork, Technical Documentation, Dashboard Development, Business Value Quantification, ROI Analysis, Presentation Skills

## WORK EXPERIENCE

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| <b>Teaching Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | August 2025 – May 2026 |
| <i>C.T. Bauer College of Business, University of Houston, Houston TX</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                        |
| <ul style="list-style-type: none"><li>Supported instructional delivery for <b>1,200+ students</b> across two semesters; maintained Excel-based grade tracking systems, applying descriptive statistics (mean, standard deviation, percentile ranking) to analyze score distributions, identify learning gaps, and deliver structured performance reports to faculty for data-driven curriculum adjustments.</li><li>Performed <b>quarterly decision auditing</b> for the SMS (Sales Management Simulation) platform, validating strategic inputs submitted by 100+ individual student teams across competitive industry groups; applied comparative analysis against industry-peer benchmarks to detect anomalous or out-of-range decision values and escalate findings to the course instructor.</li><li>Conducted <b>cross-team KPI analysis</b> on simulation performance metrics — cumulative profit, market share, Net Promoter Score (NPS), and company reputation — across 8 simulated fiscal quarters and multiple market scenarios (high-growth, early-peak, early-saturation), producing structured reports to assess student decision quality and strategic trade-off comprehension.</li><li>Analyzed student-submitted <b>sales channel allocation decisions</b> across personal selling, online, specialty retail, and in-store distribution segments, identifying behavioral patterns in product mix strategies and communicating findings to faculty to support assessment design and targeted student feedback.</li></ul> |                        |

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| <b>Data Analyst</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Jan 2023 – Jul 2024 |
| <i>BytesEdge Pvt. Ltd., Hyderabad, India</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                     |
| <ul style="list-style-type: none"><li>Performed end-to-end <b>data cleaning and preprocessing</b> on structured datasets sourced from web and mobile platform operations using Python (Pandas, NumPy); handled missing value imputation, duplicate record removal, datatype standardization, and outlier detection to ensure data integrity prior to analysis.</li><li>Wrote and optimized <b>SQL queries</b> (joins, subqueries, window functions, aggregations) to extract, transform, and load data from relational databases, supporting recurring business reporting needs and ad-hoc analytical requests from stakeholders.</li><li>Designed and maintained interactive <b>business intelligence dashboards</b> in <b>Tableau</b> and <b>Microsoft Power BI</b>, visualizing KPIs such as user engagement metrics, platform performance trends, and client conversion rates to support data-driven decision-making across business units.</li><li>Conducted <b>exploratory data analysis (EDA)</b> and <b>basic statistical analysis</b> — including descriptive statistics, correlation analysis, and trend identification — to surface actionable insights from client and operational datasets aligned with the company's digital agility and business intelligence focus.</li><li>Built and automated <b>Excel-based reporting pipelines</b> using pivot tables, VLOOKUP/XLOOKUP, and conditional formatting to deliver weekly and monthly performance summaries to project leads, reducing manual reporting effort.</li><li>Collaborated with cross-functional teams including software developers and project managers to translate business requirements into analytical frameworks, define data collection schemas, and ensure reporting outputs aligned with client deliverables.</li></ul> |                     |

## PROJECTS

### Interactive Population Visualization of Virginia | *VTK · Vega-Lite · Trame · Python*

Aug 2025 – Dec 2025

*K-Means · Linear Regression*

- Engineered a full-stack, browser-based visual analytics platform integrating VTK for polygonal county mesh rendering with Vega-Altair (compiled to Vega-Lite) for multi-view statistical charting, unified through a Trame reactive state framework that synchronizes user interactions bidirectionally across 8 analytical panels including temporal trends, population share, year-over-year growth, correlation views, and choropleth map encodings.
- Designed a modular backend architecture with dedicated chart-builder functions per analytical tab, fusing heterogeneous data sources — tabular population CSV and polygonal VTK boundary mesh — through identifier-matched joins to enable seamless integration of spatial, temporal, and statistical signals within a single interactive environment.
- Integrated lightweight ML capabilities including K-Means clustering on normalized population time-series for unsupervised demographic segmentation and per-county linear regression for predictive trend forecasting, with results surfaced directly through synchronized chart views and color-coded map encodings across dynamic county-level selections.

### Gender Classification from Blogpost Text | *Python · Scikit-learn · TF-IDF · POS Tagging*

Jan 2025 – May 2025

*PCA · CNN · BERT/Contextual Embeddings · XGBoost*

- Designed a multi-stage NLP classification pipeline on a 3,232-entry labeled blog corpus, progressing from lexical baselines using TF-IDF and CountVectorizer representations with Logistic Regression (L1/L2), SVM, and XGBoost classifiers, to syntactically enriched feature sets via Part-of-Speech (POS) tag frequency vectors and POS-stratified TF-IDF applied independently across noun, verb, adjective, and adverb sub-corpora.
- Engineered handcrafted linguistic features — including sentiment polarity, subjectivity scores, noun/verb/adjective counts, word count, and average word length — combined with TF-IDF vectors and applied Principal Component Analysis (PCA) for dimensionality reduction to mitigate the curse of dimensionality and improve generalization across Logistic Regression, SVM, Random Forest, and Naive Bayes classifiers.
- Leveraged pre-trained contextual embeddings fed into a Convolutional Neural Network (CNN) for local semantic pattern extraction, demonstrating statistically significant accuracy gains over all traditional baselines and validating the superiority of deep contextual representations over handcrafted feature engineering for stylometric gender classification.

### Lung Cancer Detection From CT Scans | *Python · PyTorch · MONAI · SimpleITK · Pandas · Scikit-learn · Jupyter*

Jan 2024 – May 2024

- Built an end-to-end lung nodule detection pipeline on real CT scans from the LUNA16 dataset (178 scans), including volumetric image ingestion, Hounsfield Unit windowing, isotropic resampling, and lung segmentation via thresholding and connected-component analysis; implemented a self-contained candidate-detection stage using local-maxima search to locate nodules directly from full scans rather than relying on pre-labeled candidates.
- Trained and benchmarked 3D CNN classifiers - a custom lightweight CNN versus MONAI's DenseNet121 - on labeled nodule/false-positive patches, applying class-weighted loss, negative subsampling, and 3D flip augmentation to address a ~400:1 class imbalance; the DenseNet3D model improved test AUC from 0.83 to 0.96 and nodule-class precision from 37% to 73% at matched ~80% recall.
- Evaluated the full detection pipeline using the clinical-standard FROC methodology (sensitivity vs. false-positives-per-scan) on a held-out test split, computing the LUNA16 CPM score across seven standard operating points; prioritized sensitivity over precision consistent with a cancer-screening use case, and identified concrete data/segmentation improvements to close the gap toward competitive benchmarks.

## CERTIFICATIONS

Applied Data Science powered by Google, *SmartInternz*

Google Cloud Computing Foundations , *Google*

Overview of Space Science, *ISRO*

May 2023 – July 2023

May 2023 – July 2023

Aug 2023 – Sept 2023